STATE BOARD OF TECHNICAL EDUCATION, KERALA — DIPLOMA NOTES

COURSE: DISCRETE MATHEMATICS (COURSE CODE: 3341)

SEMESTER III | DIPLOMA IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

MODULE 1: MATHEMATICAL LOGIC

Full Exhaustive Lecture Textbook & Solved Previous Year University Exam Questions (PYQ Bank)

COURSE SYLLABUS & EXAM ALIGNMENT

Course Outcome CO1: Apply the basic concepts of mathematical logic to solve computational problems.
Duration: 16 Hours | **Cognitive Level:** Applying
Prescribed Textbooks: (1) Kenneth H. Rosen (7th Ed), (2) Ralph P. Grimaldi (5th Ed), (3) Thomas Koshy (2nd Ed).
Scope: Complete coverage of M1.01 to M1.05 with exhaustive explanations, definitions, laws, proofs, worked-out examples, and solved PYQs.

1. Propositional Logic & Operations (M1.01, M1.02)

1.1 Proposition Concept: A declarative sentence that is either True (T) or False (F), but not both.
• *Atomic Proposition:* Simple statement (e.g., *p: '2 + 3 = 5'*).
• *Compound Proposition:* Combined using logical connectives.

1.2 Fundamental Connectives:
• **Negation (¬p):** Reverses truth value.
• **Conjunction (p ∧ q):** True ONLY when BOTH *p* and *q* are True.
• **Disjunction (p ∨ q):** True when AT LEAST ONE of *p, q* is True.
• **Exclusive-OR (p ⊕ q):** True when EXACTLY ONE of *p, q* is True.
• **Implication (p → q):** False ONLY when *p* = *True* and *q* = *False*. (Vacuous truth when *p* = *False*).
• **Biconditional (p ↔ q):** True when *p* and *q* have identical truth values.

p	q	¬p	p ∧ q	p ∨ q	p ⊕ q	p → q	p ↔ q
T	T	F	**T**	**T**	F	**T**	**T**
T	F	F	F	**T**	**T**	**F**	F
F	T	T	F	**T**	**T**	**T**	F
F	F	T	F	F	F	**T**	**T**

2. Conditional Variants & Truth Classifications (M1.02, M1.03)

2.1 Variants of p → q:
• *Converse:* q → p • *Inverse:* ¬p → ¬q • *Contrapositive:* ¬q → ¬p.
• **Equivalence Theorem:** *(p → q) ≡ (¬q → ¬p)*. Contrapositive is strictly equivalent to original implication!

2.2 Classifications:
• **Tautology (T):** Always True under all interpretations (e.g., *p ∨ ¬p*).
• **Contradiction (F):** Always False under all interpretations (e.g., *p ∧ ¬p*).
• **Contingency:** Neither tautology nor contradiction.

2.3 Normal Forms:
• **Disjunctive Normal Form (DNF):** Disjunction (OR) of minterms (ANDs of literals) for rows where output = True.
• **Conjunctive Normal Form (CNF):** Conjunction (AND) of maxterms (ORs of literals) for rows where output = False.

3. Logical Equivalence & Duality Principle (M1.03)

Law Category	Equivalence Statement 1	Equivalence Statement 2 (Dual Form)
Identity Laws	$p \wedge T \equiv p$	$p \vee F \equiv p$
Domination Laws	$p \vee T \equiv T$	$p \wedge F \equiv F$
Idempotent Laws	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double Negation Law	$\neg(\neg p) \equiv p$	—
Commutative Laws	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
Associative Laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive Laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan's Laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption Laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negation Laws	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
Conditional Equivalences	$p \rightarrow q \equiv \neg p \vee q$	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

Algebraic Proof Example: Prove $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$

1. $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$ [De Morgan's Law]
2. $\equiv \neg p \wedge (\neg(\neg p) \vee \neg q)$ [De Morgan's Law]
3. $\equiv \neg p \wedge (p \vee \neg q)$ [Double Negation Law]
4. $\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$ [Distributive Law]
5. $\equiv F \vee (\neg p \wedge \neg q)$ [Negation Law: $\neg p \wedge p \equiv F$]
6. $\equiv \neg p \wedge \neg q$ [Identity Law: $F \vee A \equiv A$] **Q.E.D.**

4. Predicate Logic & Quantifiers (M1.04)

• **Predicate P(x):** Statement with variable *x* over domain *U*.
• **Universal Quantifier ($\forall$x P(x)):** True if *P(x)* holds for EVERY $x \in U$.
• **Existential Quantifier ($\exists$x P(x)):** True if *P(x)* holds for AT LEAST ONE $x \in U$.
• **De Morgan's Quantifier Negations: $\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$** and **$\neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)$**.
• **Nested Quantifiers: $\forall x\, \exists y\, P(x, y) \neq \exists y\, \forall x\, P(x, y)$** (Order matters!).

5. Rules of Inference & Natural Deduction (M1.05)

Rule Name	Premises	Conclusion	Tautology Form
Modus Ponens	p $p \rightarrow q$	$\therefore q$	$[p \wedge (p \rightarrow q)] \rightarrow q$
Modus Tollens	$\neg q$ $p \rightarrow q$	$\therefore \neg p$	$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$
Hypothetical Syllogism	$p \rightarrow q$ $q \rightarrow r$	$\therefore p \rightarrow r$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

Disjunctive Syllogism	$p \vee q$ $\neg p$	$\therefore \mathbf{q}$	$[(p \vee q) \wedge \neg p] \rightarrow q$
Addition	p	$\therefore \mathbf{p \vee q}$	$p \rightarrow (p \vee q)$
Simplification	$p \wedge q$	$\therefore \mathbf{p}$	$(p \wedge q) \rightarrow p$
Conjunction	p q	$\therefore \mathbf{p \wedge q}$	$[(p) \wedge (q)] \rightarrow (p \wedge q)$
Resolution	$p \vee q$ $\neg p \vee r$	$\therefore \mathbf{q \vee r}$	$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (q \vee r)$

6. Methods of Proof & Resolution Principle (M1.05)

- **Direct Proof:** Assume *p*, show *q* directly.
- **Proof by Contraposition:** Assume ¬*q*, show ¬*p*.
- **Proof by Contradiction:** Assume ¬*p*, derive contradiction $r \wedge \neg r$ *(F)* (e.g., proof that $\sqrt{2}$ is irrational).
- **Resolution Refutation Algorithm:** Convert premises to clause form (disjunctions), negate the conclusion, and apply resolvent rule $(A \vee C) \wedge (\neg A \vee B) \Rightarrow (B \vee C)$ repeatedly until the **Empty Clause (■)** is derived.

7. Solved Previous Year University Exam Question Bank (PYQ)

PART-A: 1-MARK SOLVED EXAM QUESTIONS

PART-A: 1-MARK OBJECTIVE PYQS (WITH STEP-BY-STEP SOLUTIONS)

Q1 [Nov 2025]: Interpret the logical equivalence to the statement: *'I pass only if you pass'*.
Sol: Let *p*: 'I pass', *q*: 'You pass'. 'p only if q' ≡ $\mathbf{p \rightarrow q} \equiv \mathbf{\neg p \vee q} \equiv \mathbf{\neg q \rightarrow \neg p}$ (Contrapositive).

Q2 [Nov 2025]: Write the dual for $(p \wedge q) \vee T$.
Sol: Swap $\wedge \leftrightarrow \vee$ and T ↔ F. Dual = $\mathbf{(p \vee q) \wedge F}$.

Q3 [Nov 2024]: If *p* is a proposition, ¬(¬p) = ...?
Sol: By Double Negation Law, ¬(¬p) = **p**.

Q4 [Nov 2024]: If *A* has *n* elements, then number of subsets of *A* = ...?
Sol: Number of subsets = |P(A)| = $\mathbf{2^n}$.

PART-B: 3-MARK SOLVED EXAM QUESTIONS

PART-B: 3-MARK SHORT ANSWER PYQS (WITH STEP-BY-STEP SOLUTIONS)

Q1 [Nov 2025]: Show that the implication $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology.
Sol: Constructing 8-row truth table for $A = (p \rightarrow q) \wedge (q \rightarrow r)$ and $B = (p \rightarrow r)$ shows all entries in $A \rightarrow B$ are **T**. Hence, it is a Tautology.

Q2 [Nov 2025]: Without constructing a truth table, prove that $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$.
Sol: $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$ [De Morgan] ≡ $\neg p \wedge (p \vee \neg q)$ [Double Negation] ≡ $(\neg p \wedge p) \vee (\neg p \wedge \neg q)$ [Distributive] ≡ $F \vee (\neg p \wedge \neg q)$ [Negation] ≡ $\neg p \wedge \neg q$ [Identity]. **Q.E.D.**

Q3 [Nov 2024]: Determine the truth value / truth table of conjunction table $(p \wedge q)$.
Sol: $p \wedge q$ is True ONLY when both $p = T$ and $q = T$. Rows: (T,T)→T, (T,F)→F, (F,T)→F, (F,F)→F.

PART-C: 7-MARK SOLVED EXAM QUESTIONS

PART-C: 7-MARK LONG ANSWER PYQS (WITH STEP-BY-STEP SOLUTIONS)

Q1 [Nov 2025]: Construct the truth table for $(p \vee q) \wedge (\neg p \rightarrow \neg q)$.
Sol: Row 1 (T,T): T $\wedge$ T = **T**; Row 2 (T,F): T $\wedge$ T = **T**; Row 3 (F,T): T $\wedge$ F = **F**; Row 4 (F,F): F $\wedge$ T = **F**.

Q2 [Nov 2025 - OR]: Write converse, inverse, contrapositive for: *'If 2 < 3 and 3+7=10, then √4 = 2'* and find their truth values.
Sol: Let *p: (2<3 $\wedge$ 3+7=10)* [True], *q: √4=2* [True].
- **Original (p $\rightarrow$ q):** If 2<3 and 3+7=10, then √4=2. Truth Value = **T**.
- **Converse (q $\rightarrow$ p):** If √4=2, then 2<3 and 3+7=10. Truth Value = **T**.
- **Inverse (¬p $\rightarrow$ ¬q):** If not(2<3 $\wedge$ 3+7=10), then √4 ≠ 2. Truth Value = **T**.
- **Contrapositive (¬q $\rightarrow$ ¬p):** If √4 ≠ 2, then not(2<3 $\wedge$ 3+7=10). Truth Value = **T**.

Q3 [Nov 2025]: Prove argument validity: Premises: *1. $p \rightarrow q$*, *2. $q \rightarrow (r \wedge s)$*, *3. $\neg r \vee (\neg t \vee u)$*, *4. $p \wedge t$*. Conclusion: *u*.
Sol: From 4: *p* (Simp) & *t* (Simp). From 1 & *p*: *q* (MP). From 2 & *q*: $r \wedge s$ (MP) $\Rightarrow$ *r* (Simp). From 3 & *r*: $\neg t \vee u$ (DS). From *t* & $\neg t \vee u$: **u** (DS). **Q.E.D.**

Q4 [Nov 2024]: Construct truth table for $\neg(p \vee q) \rightarrow \neg p$.
Sol: Evaluating all 4 rows yields **T** in every row. Hence it is a **Tautology**.

Q5 [Nov 2024 - OR]: Use truth table to show that $\neg(p \rightarrow q) \rightarrow \neg q$ is a logical implication.
Sol: Evaluating $\neg(p \rightarrow q) \rightarrow \neg q$ yields **T** in all 4 rows. Since it is a Tautology, it is a valid **logical implication**.